While problem of bias and discrimination have always been present in society, the worry is that AI may perpetuate these problems and enlarge their impact.

Potential Harms Caused by AI Systems

COMPAS: Racial Bias in Criminal Justice Systems

Bernard Parker, left, was rated high risk; Dylan Fugett was rated low risk. (Josh Ritchie for ProPublica)

Machine Bias

There's software used across the country to predict future criminals. And it's biased against blacks.

by Julia Angwin, Jeff Larson, Surya Mattu and Lauren Kirchner, ProPublica

May 23, 2016

https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing

Description of COMPAS

Correctional Offender Management Profiling for Alternative Sanctions

- Intends to provide **risk-assessments** for individuals.
- These assessments were meant **to measure the likelihood that someone may re-offend** (recidivism), whether someone may go on to commit a violent crime.
- As the algorithm primarily concerns itself with risk of recidivism, it is **primarily meant to be used in making decisions of pretrial release (bail) and parole**.

COMPAS Risk Assesment Survey:
https://embed.documentcloud.org/documents/2702103-Sample-Risk-Assessment-COMPAS-CORE/?mode=document&embed=1

Practitioner's Guide to COMPAS Core: https://s3.documentcloud.org/documents/2840784/Practitioner-s-Guide-to-COMPAS-Core.pdf

Incident 1:

- After school in 2014, two 18-year-old girls, Brisha Borden and Sade Jones, running late to pick up her sister, briefly grabbed an unlocked bicycle and scooter.
- But realized they were too big for the small items, which belonged to a 6-year-old.
- The act seemed impulsive to a neighbour.
- Police arrived and arrested the girls for burglary and theft of $80 worth of goods.

Incident 2:

- A 41-year-old man, Vernon Prater, was caught shoplifting $86 worth of goods from Home Depot.
- He had prior convictions for armed robbery and had previously served five years in prison.

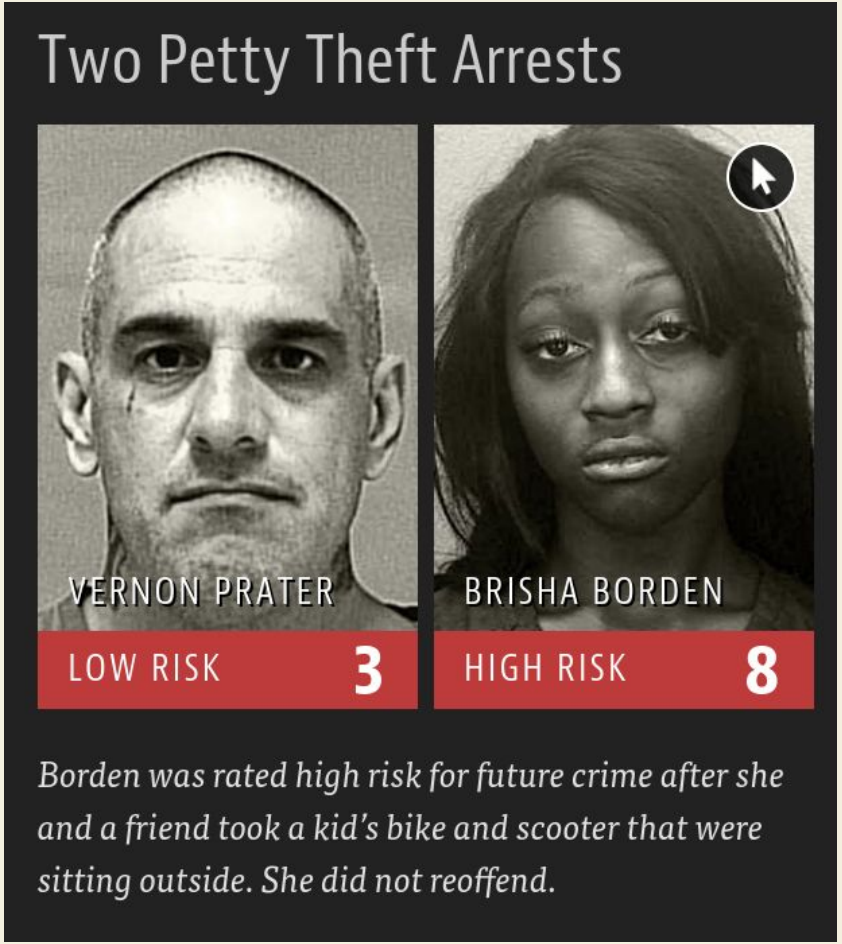
- Both groups were booked into jail, where a judge decides how to set bail:

  should they be released from jail, with some amount of money as collateral, while they await trial?

- Broward County uses COMPAS, to assist the judge in making such decisions.

- Two years later,

- Borden has not been charged with any new crimes.
- Prater is serving an eight-year prison term for subsequently breaking into a warehouse and stealing thousands of dollars' worth of electronics.



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Description of COMPAS

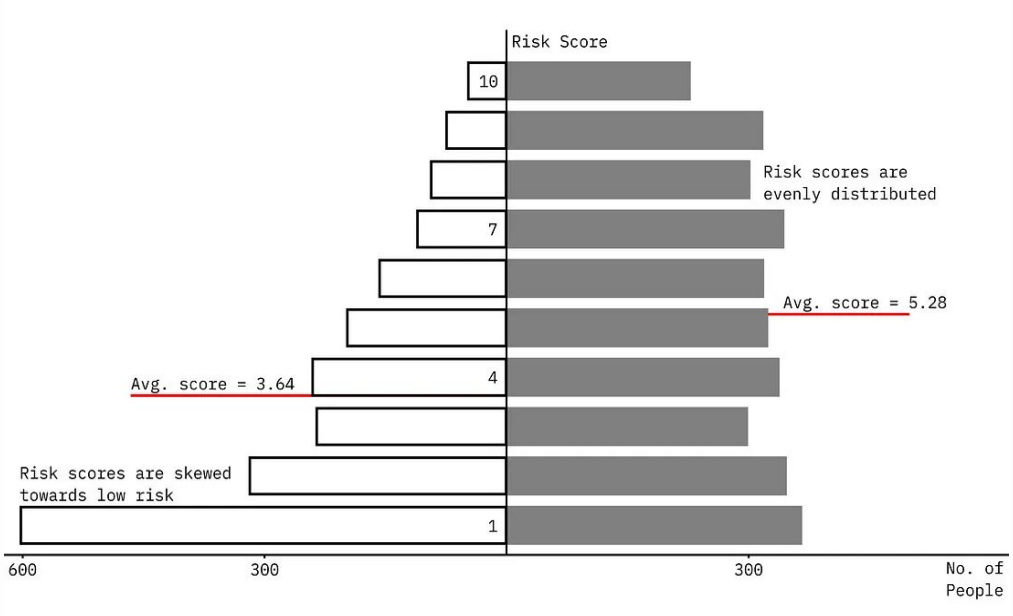
Once the score became available to the criminal justice system as a whole, it became used in wholly inappropriate ways.

“.... as identified, through the COMPAS assessment, as an individual who is at high risk to the community. …. ” *- A judge during a sentencing decisions.*

Whether someone may commit another crime in the future, does that has a bearing on whether they did or did not commit the current crime *in question?*

Propublica’s analysis

- a non-profit investigative journalism newsroom, performed a study of the Risk scores assigned to 7000 people arrested in Broward County, Florida in 2013 and 2014.
- From their findings they claimed that the COMPAS algorithm was biased against Black arrestees.

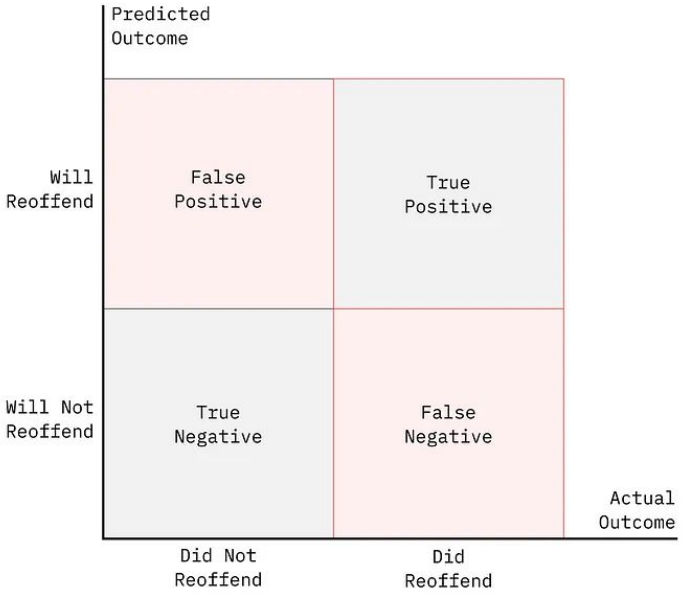


https://medium.com/@lamdaa/compas-unfair-algorithm-812702ed6a6a

ProPublica github repository, including dataset: https://github.com/propublica/compas-analysis/

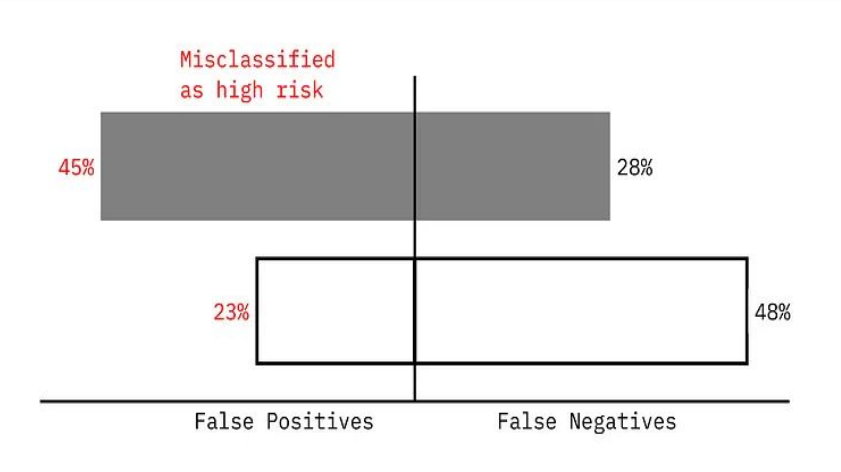
Difference in False positive & False negative rates

As ProPublica takes a binary approach to evaluate the algorithm's performance, (whether a prediction was true of false), we get four possible scenarios:



Difference in False positive & False negative rates

- The errors were in favour of Whites. The errors were unfavourable towards Blacks.
- Black defendants were twice as likely to be misclassified as high risk. White defendants were more likely to be misclassified as low risk.

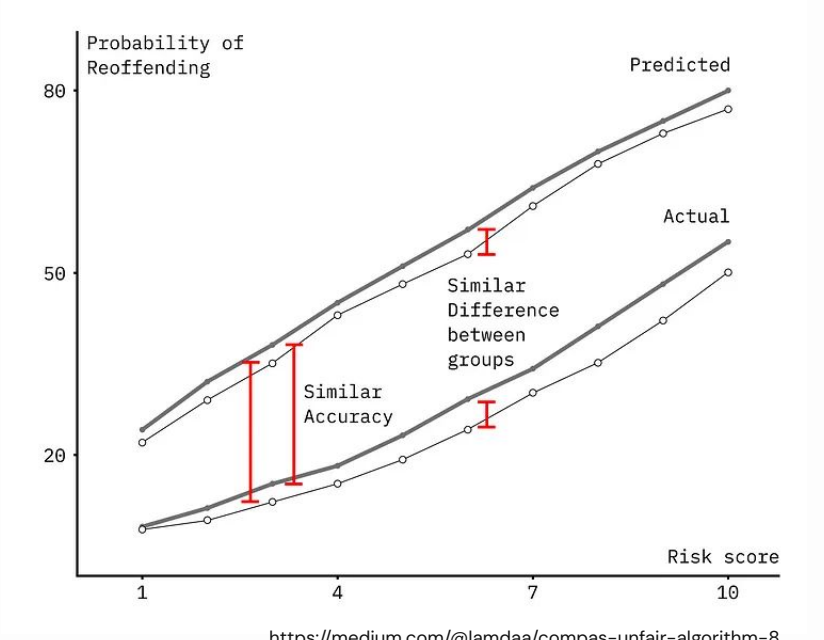


But Risk scores predict probability of Recidivism equally accurately

- Accuracy of the predicted probability is similar for both populations.
- The differences in predicted probability for both groups reflects actual probabilities.

  how the results were biased in some ways but unbiased in other ways?

- NorthPointe (now Equivant), the company that owns COMPAS argued that COMPAS is equally accurate at predicting risk of future recidivism for Black & White defendants, and therefore not biased.



https://medium.com/@lamdaa/compas-unfair-algorithm-812702ed6a6a

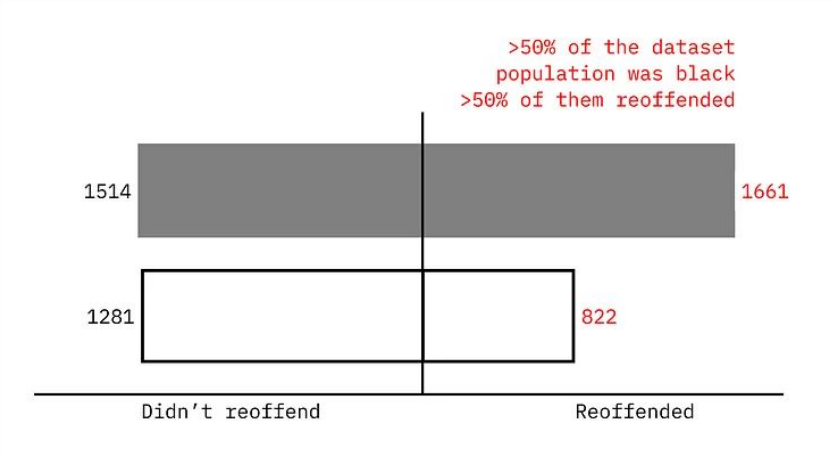
Northpointe response to ProPublica: https://go.volarisgroup.com/rs/430-MBX-989/images/ProPublica_Commentary_Final_070616.pdf

Who is correct here ?

Bias in:

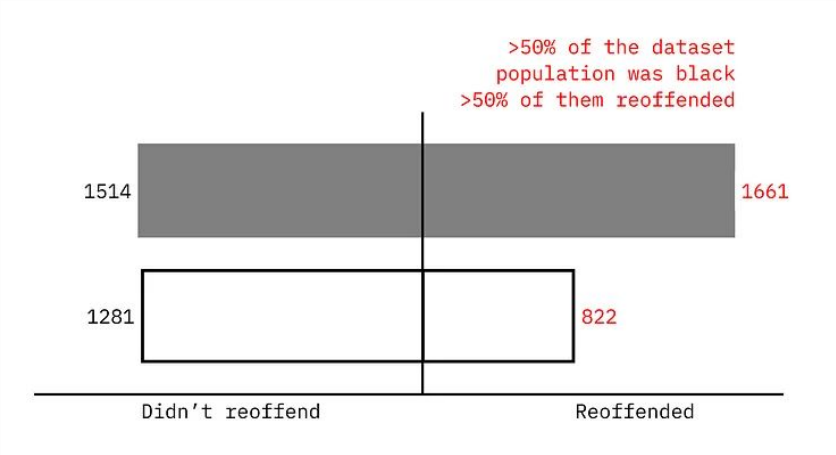
1. The probability of actual recidivism was higher for Black defendants.
2. The total number of recidivists for every risk category (Low-Med-High) was higher for Black arrestees.

Greater number of people in the dataset were Black. They also re-offended at a higher rate.



Bias in, Bias out

“If we’re living in a reality that is biased, an algorithm that is designed to be able to accurately predict this reality, will end up making biased predictions.”



Bias in the future as a result of bias in the past ?

Algorithmic systems can help us address the current biases in the criminal justice system.

Should justice be blind and impartial, or does justice mean creating an advantage for those who are already disadvantaged?

Should the data scientist make the training data an “idealized” data?